ELECTRIC VEHICLE MARKET PERFORMANCE & INSIGHTS

Project Report

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Live website: https://ev-market-intelligence.onrender.com/

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1. INTRODUCTION

1.1 Project Overview

EV Market Intelligence is a descriptive analytics project that turns a structured electric-vehicle dataset into a set of SQL summaries, Tableau visualizations, an interactive dashboard, a three-scene Tableau Story and a Flask web page. It is intended to make comparisons across EV categories, manufacturers, manufacturing countries, market regions, model years and battery technologies easier to explore.

Important scope note: the supplied dataset is procedurally generated and its manufacturer/model names are fictional. The visualizations describe patterns in this synthetic dataset and must not be presented as verified real-world EV market rankings or production statistics.

1.2 Purpose

The purpose is to demonstrate an end-to-end workflow: collect a course-linked dataset, inspect and prepare records, store/query data with SQLite, create descriptive Tableau charts and narrative scenes, publish the views, and embed them in a Flask website that can be opened through a browser.

Project facts	Value
Project title	Electric Vehicle Market Performance & Insights
Author	Prasiddha Khanna (individual project)
Team ID	N/A - not provided
Analysis records	199,999 cleaned rows; 37 columns
Web demo	https://ev-market-intelligence.onrender.com/
Tableau dashboard	https://public.tableau.com/shared/Q27K39PSC?:display_count=n&:origin=viz_share_link
Tableau Story	https://public.tableau.com/views/EV_Market_Intelligence/Story1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

2. IDEATION PHASE

2.1 Problem Statement

A learner or analyst exploring an EV dataset may need to compare recorded sales and vehicle attributes across categories, manufacturers, geography, model years and battery technologies. When the measures are reviewed separately, the comparisons can require repeated manual work. The proposed solution consolidates the analysis into a dashboard, story and web portal. This is a project framing statement, not a confirmed finding from user interviews.

2.2 Empathy Map Canvas

Persona hypothesis: a student or analyst who needs to compare category demand, battery characteristics and geography. No formal user interviews or surveys were conducted; therefore, the Says/Thinks/Does/Feels entries in the associated empathy-map template are explicit design assumptions, not reported quotations.

Says	Thinks	Does	Feels
Needs fast comparison across categories and battery types.	Wonders which patterns are meaningful and how the figures were generated.	Filters the dashboard, compares visual encodings, checks tooltips and opens the three Story scenes.	May feel overloaded by raw fields before analysis and may find a structured view easier to scan; this is a hypothesis and was not user-tested.

2.3 Brainstorming

The solution options considered in the documentation were an interactive category/region dashboard, a battery-technology comparison, a three-scene Story and a Flask web portal embedding the published views. The

dashboard plus Story plus web integration was selected because it combines comparisons with a shareable narrative and deployable interface. The ideation record is retrospective; it should not be read as a multi-member workshop.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

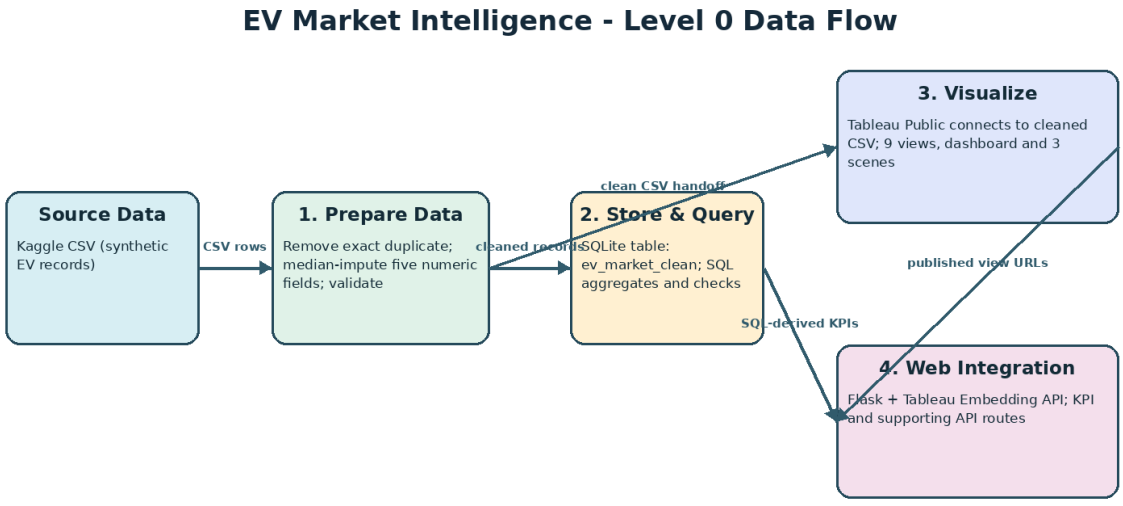
Stage	User action	Touchpoint	Question / thought	Improvement opportunity
Discover need	Identify a comparison question about category, geography or battery technology.	Flask landing page and dashboard title.	“Where should I start?”	Keep titles, KPI definitions and chart purposes obvious.
Explore and compare	Use region/category/year/battery filters and compare charts.	Tableau marks, filters and tooltips.	“How do the groups differ?”	Apply filters consistently and label aggregations clearly.
Interpret and share	Navigate the Story and share a public view or website link.	Three-scene Story and Flask page.	“What can I safely conclude?”	Emphasize that the dataset is synthetic and findings are descriptive.

3.2 Solution Requirements

- Functional: load the cleaned CSV and SQLite dataset, calculate descriptive summaries, support grouping by category/manufacturer/region/model year/country/battery type, render nine views, supply four dashboard filters, show the Story, and embed published views in Flask.
- Non-functional: readable labels, responsive embedding, reproducible setup, no committed secrets, clear synthetic-data warning and a health endpoint. Exact response/render timing targets have not been measured.

3.3 Data Flow Diagram

The data-preparation script creates the cleaned analysis file and the SQLite database. SQL queries run against SQLite for data checks and grouped outputs. Tableau Public uses the prepared CSV connection because the supplied setup does not establish direct SQLite connectivity in Tableau Public. The Flask application embeds the published Tableau views and reads the runtime database for API outputs where configured.



3.4 Technology Stack

Layer	Technology	Use
Preparation	Python / pandas	Deduplicate and impute numeric missing values in supplied preparation workflow.
Database	SQLite	Table ev_market_clean and SQL queries/data checks.
Visualization	Tableau Public Desktop / Tableau Public	Nine charts, filters, dashboard and three Story scenes using cleaned CSV.
Web app	Flask, HTML, CSS, JavaScript, Tableau Embedding API v3	Dashboard/Story embed and API endpoints.
Hosting/source	Render and GitHub	Public web deployment and source repository; confirm repository visibility/contents before hand-in.

4. PROJECT DESIGN

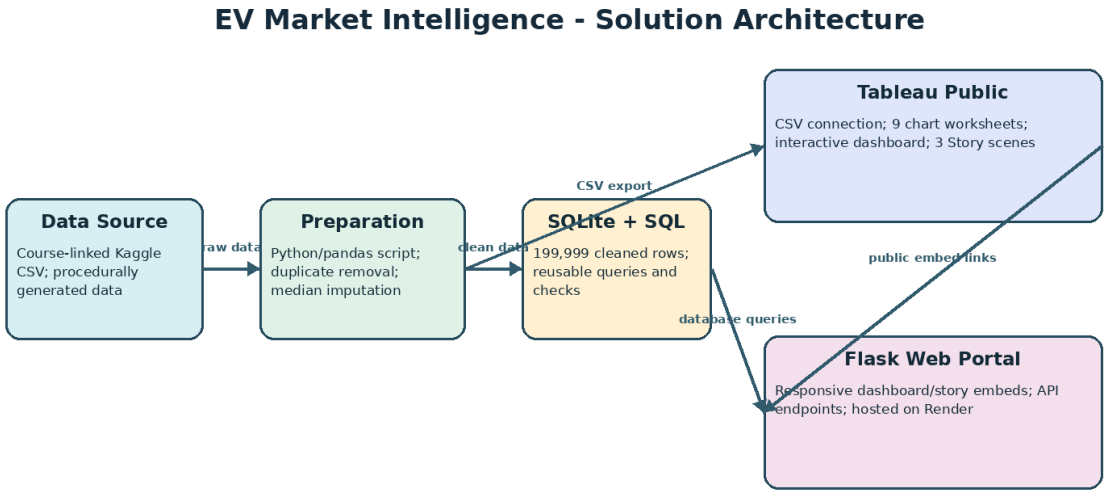
4.1 Problem-Solution Fit

Solution-fit hypothesis: a dashboard plus three-scene Story should reduce the effort needed to compare the supplied dataset by placing relevant fields and aggregations into dedicated visualizations. No formal user research or usability study was performed; benefit claims are intended goals rather than measured outcomes.

4.2 Proposed Solution

The solution uses a cleaned CSV for Tableau Public, a SQLite database and SQL query pack for database analysis, nine distinct chart worksheets, one interactive dashboard, a three-scene Story and a Flask app hosted at Render. It provides descriptive comparisons rather than a predictive model or validated market forecasting.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

The work was completed as an individual project over an approximate September-October 2026 timeline, as supplied by the project owner. The sprint-style entries in the associated planning template are retrospective estimates for documentation; no formal daily burndown or measured team velocity was maintained.

Work phase	Approximate period	Outputs
Problem and data review	September 2026	Problem framing, dataset inspection and

		field understanding.
Preparation and SQL	September 2026	Cleaned CSV, SQLite table and SQL query pack.
Tableau visualization	September-October 2026	Nine chart worksheets, dashboard filters and three Story scenes.
Web integration and publication	October 2026	Flask dashboard/Story embeds, Render website and documentation templates.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Data Quality and Database Validation

Check	Evidence / result
Rows before cleaning	200,000 training rows reported in supplied data-quality report.
Rows after cleaning	199,999 rows; 37 columns.
Duplicate handling	One exact duplicate row / vehicle ID removed.
Missing values before cleaning	18,996 missing cells across five numeric columns.
Imputation	Median imputation for battery_cycle_life (2031.0), resale_value_pct_3yr (52.99), infotainment_screen_inch (11.0), annual_sales_units (3949.0) and customer_rating (4.1).
Distinct vehicle IDs	199,999 distinct IDs in SQLite; equal to cleaned row count.
Invalid-range checks	Supplied query pack checks negative annual sales and nonpositive range; direct query output showed 0 rows for those two conditions.

6.2 Performance Testing

The Tableau performance template was populated with verifiable configuration facts: 199,999 records, four dashboard filters, five calculated-field definitions in the build guide, nine visualizations and three Story scenes. Exact Tableau render times, filter latencies, query-history timings and measured user-load tests were not recorded, so they are explicitly marked “not measured” in the performance template rather than fabricated.

6.3 Web Route Checks

An Antigravity execution report previously stated 5/5 local checks passed: /health, /api/kpis, /api/region-sales, /api/battery-tech and homepage. This report is retained as agent-reported evidence; it was not independently rerun while creating this documentation package. The live demo URL was provided by the project owner.

7. RESULTS

The SQLite database verification file reports 199,999 rows and distinct vehicle IDs, total recorded annual sales units of 1,558,703,498, average range of 324.5 km, average price of USD 64,921, average customer rating of about 4.09 and seven market regions. The dataset is synthetic, so these are aggregates of generated data only.

7.1 Output Screenshots and Visualizations

Visualization	Question / encoding
1 - EV Category Performance & Distribution	Bar chart: sum of annual sales units by category.
2 - Customer Rating Records by EV Category	Packed bubbles: count of customer-rating records by category.
3 - Market Share by Manufacturer	Treemap: sum of recorded annual sales units by manufacturer.
4 - Annual EV Sales Trend by Model Year	Line chart: sum of recorded sales units by model year, 2018-2026.
5 - Battery Degradation & Efficiency Matrix	Heat map: category by battery type, color by average degradation metric used in workbook.
6 - Battery Mix	Pie chart: recorded sales by battery type.
7 - EV Distribution by Manufacturing Country	Filled map: recorded sales by country.
8 - Driving Range vs Battery Capacity	Scatter plot: average battery capacity versus range, colored by battery type.
9 - Average Price by Vehicle Category	Dot plot: average price by category.

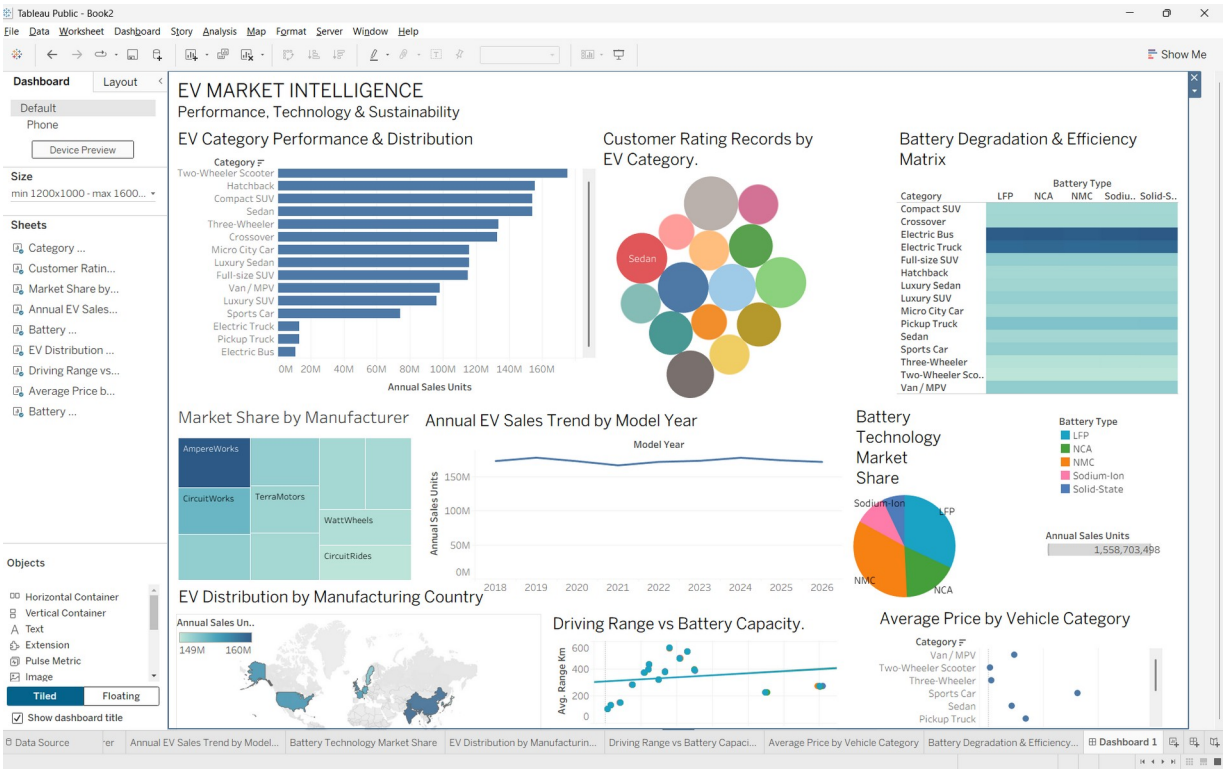


Figure 1. Tableau Desktop dashboard screenshot supplied during project work; published dashboard link is listed in the Appendix.

7.2 Descriptive Sample Findings

- Highest category total: Two-Wheeler Scooter, 175,389,726 recorded sales units; followed by Hatchback (155,589,242) and Compact SUV (154,153,576).
- Highest manufacturer total: AmpereWorks, 9,504,546 recorded sales units. These fictional manufacturer names are part of the synthetic dataset.
- Battery-type totals: NMC 526,947,105; LFP 498,490,599; NCA 268,496,582; Sodium-Ion 154,156,827; Solid-State 110,612,385 recorded sales units.
- The top manufacturing-country totals in the synthetic sample include India (159,957,989) and China (159,355,406), followed by Vietnam (158,066,930). These should not be interpreted as actual national EV production data.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Combines data preparation, SQLite querying, Tableau analytics and Flask integration into one demonstrable workflow.
- Provides different chart types matched to category, manufacturer, battery, geography and range/price questions.
- Four filters and three Story scenes support exploration and narrative explanation.
- The SQL query pack and data-quality report improve reproducibility and transparency.

Disadvantages and Limitations

- The dataset is synthetic and uses fictional manufacturer/model names; findings do not represent verified market facts.

- Five numeric columns required median imputation, which can affect their distributions; sensitivity analysis was not performed.
- Tableau Public uses the cleaned CSV connection rather than a verified direct live SQLite connection in this setup.
- Formal user research, usability testing, Tableau render-time measurements and a recorded burndown/velocity history are unavailable.
- Tableau Public views and their underlying data are public once published.

9. CONCLUSION

EV Market Intelligence demonstrates an end-to-end descriptive analytics workflow for a synthetic EV dataset. The work includes a documented cleaning pipeline, SQLite storage and SQL queries, nine distinct Tableau visualizations, a filterable dashboard, a three-scene Story and a Flask website embedding the published views. The deliverable is useful as a student analytics prototype, while its conclusions remain limited by the synthetic data and the lack of formal user/performance testing.

10. FUTURE SCOPE

- Replace or supplement synthetic data with a validated public dataset and retain source/version metadata.
- Add a field-level data dictionary, complete cleaning logs and sensitivity analysis for median imputation.
- Run measured performance testing for dashboard rendering and filter response, recording hardware, browser and repeat trials.
- Conduct user interviews and usability tests, then update the empathy map and requirements using actual feedback.
- Improve mobile layout and add more accessible descriptions/tooltips.
- Create scheduled data refresh only if data licensing, update cadence and Tableau publishing requirements permit it.
- Add role-based access controls for any future non-public data; do not treat Tableau Public as a private hosting option.

11. APPENDIX

Item	Link / location
Flask website / Demo Link	https://ev-market-intelligence.onrender.com/
Tableau Dashboard	https://public.tableau.com/shared/Q27K39PSC?:display_count=n&:origin=viz_share_link
Tableau Story	https://public.tableau.com/views/EV_Market_Intelligence/Story1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link
GitHub repository (reported; verify it is public and contains the final files)	https://github.com/PrasiddhKhanna/EV-Market-Intelligence
Dataset	Kaggle dataset link was provided on SkillWallet; copy the exact URL from the portal before final submission.
Source code / query pack	flask_app/app.py; flask_app/templates/index.html; sql/ev_market_queries.sql; scripts/prepare_ev_data.py in project repository.
Database verification	199,999 rows; 199,999 distinct vehicle IDs; total recorded annual sales 1,558,703,498.
Testing evidence	Agent-reported local Flask checks: 5/5 passed; performance latency numbers not measured.

Template compliance note: this report follows the 11-section outline in the supplied “Final Report Template.pdf”. The original template is an outline/checklist, not a fillable project-specific report; this report expands the headings with project-specific content.